

Esercizio.

Suppose B b.a. and M B -valued model. Then $\forall \phi(x) \in L \forall \sigma, \tau \in M ([[\sigma = \tau]]) \wedge [[\phi(\sigma)]] \leq [[\phi(\tau)]]$.

Dimostrazione.

- ϕ atomic: directly from definition
- $\phi : \psi \wedge \theta$: $[[\sigma = \tau]] \wedge [[\psi(\sigma)]] \wedge [[\theta(\sigma)]] \leq ([[\sigma = \tau]] \wedge [[\psi(\sigma)]]) \wedge ([[\sigma = \tau]] \wedge [[\theta(\sigma)]]) \leq [[\phi(\sigma)]]$
- $\phi : \neg \psi$: $[[\psi(\tau)]] = ([[\tau = \sigma]] \wedge [[\psi(\tau)]]) \vee (\neg [[\tau = \sigma]] \wedge [[\psi(\tau)]]) \leq [[\psi(\sigma)]] \vee (\neg [[\tau = \sigma]] \wedge [[\psi(\tau)]]) \leq [[\psi(\sigma)]] \vee \neg [[\tau = \sigma]]$
- $\phi : \exists x \psi(x, y)$: $[[\sigma = \tau]] \wedge [[\exists x \phi(x, \sigma)]] = [[\sigma = \tau]] \wedge \bigvee_{\eta \in M} [[\psi(\eta, \sigma)]] = \bigvee_{\eta \in M} [[\sigma = \tau]] \wedge [[\psi(\eta, \sigma)]] \leq \bigvee_{\eta \in M} [[\psi(\eta, \tau)]] = [[\exists x \phi(x, \tau)]]$.

□